New Science

Revise this chapter together

Unit 1 About the Earth
2 Movement of the Earth

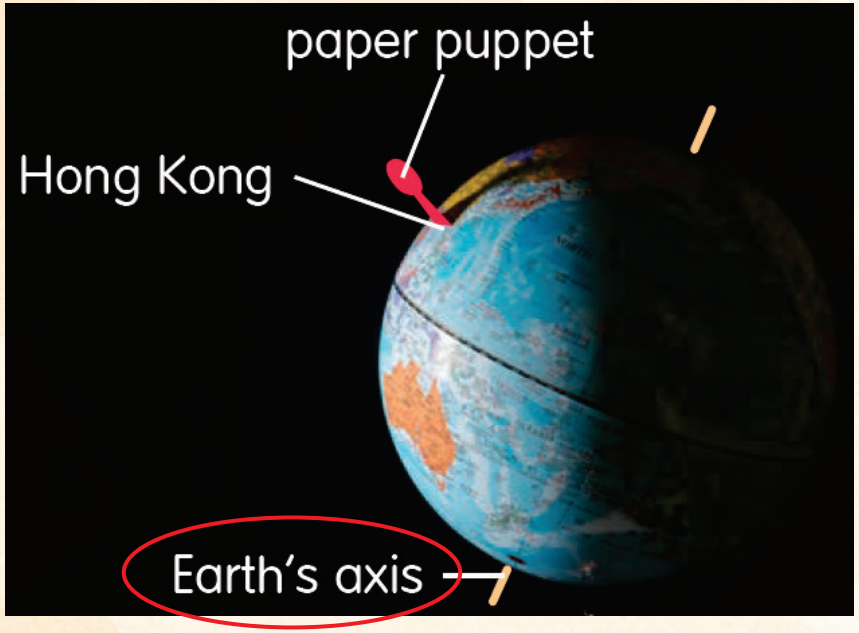




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1. The Earth rotates on its axis, which is an imaginary line passing through the North Pole and the South Pole, every day.

What does the Earth **rotate** around?

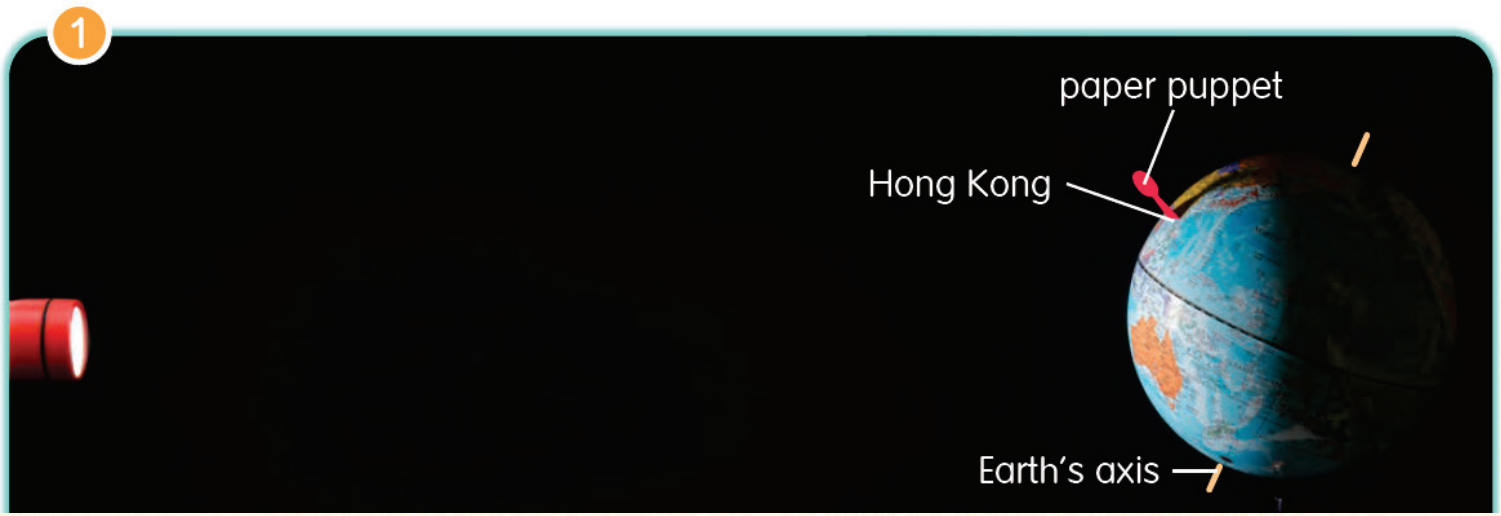


Earth's axis :

an imaginary line passes through the North Pole and the South Pole

2. It is day on the side of the Earth facing the Sun and night on the side facing away from the Sun. The rotation of the Earth causes the cycle of day and night.

Use the following way to simulate the Earth's rotation. Then, explain **why there is day and night**.



Use some tape to fix the paper puppet at the location of Hong Kong on the globe. Shine the torch on that spot.

2. It is day on the side of the Earth facing the Sun and night on the side facing away from the Sun. The rotation of the Earth causes the cycle of day and night.

Use the following way to simulate the Earth's rotation. Then, explain **why there is day and night**.

2



Without moving the torch, rotate the globe anticlockwise for a half turn.

It is bright / dark in Hong Kong when it faces the Sun, so it is day / night .

2. It is day on the side of the Earth facing the Sun and night on the side facing away from the Sun. The rotation of the Earth causes the cycle of day and night.

Use the following way to simulate the Earth's rotation. Then, explain **why there is day and night**.

3



Rotate the globe anticlockwise for another half turn.

When the globe rotates for a half turn, Hong Kong turns away from the Sun. It is bright / dark there, so it is day / night.

2. It is day on the side of the Earth facing the Sun and night on the side facing away from the Sun. The rotation of the Earth causes the cycle of day and night.



Use the following way to simulate the Earth's rotation. Then, explain **why there is day and night**.



Quick Check

The cycle of day and night is caused by the rotation of the Earth.

3. The Earth complete a full rotation in 1 day, which is about 24 hours.



Visit the Hong Kong Observatory website to find out **the time of sunrise** in Hong Kong in the past two days and find out how much time had passed between one sunrise and the next.



Quick Check

The Earth needs _24_ hours to complete a full rotation, which is _1_ day.

4. During the day, the length and direction of the shadows formed in the sunlight change at different times.

Do this activity to make a sundial and find out how its shadow changes in the sunlight at different times of the day.

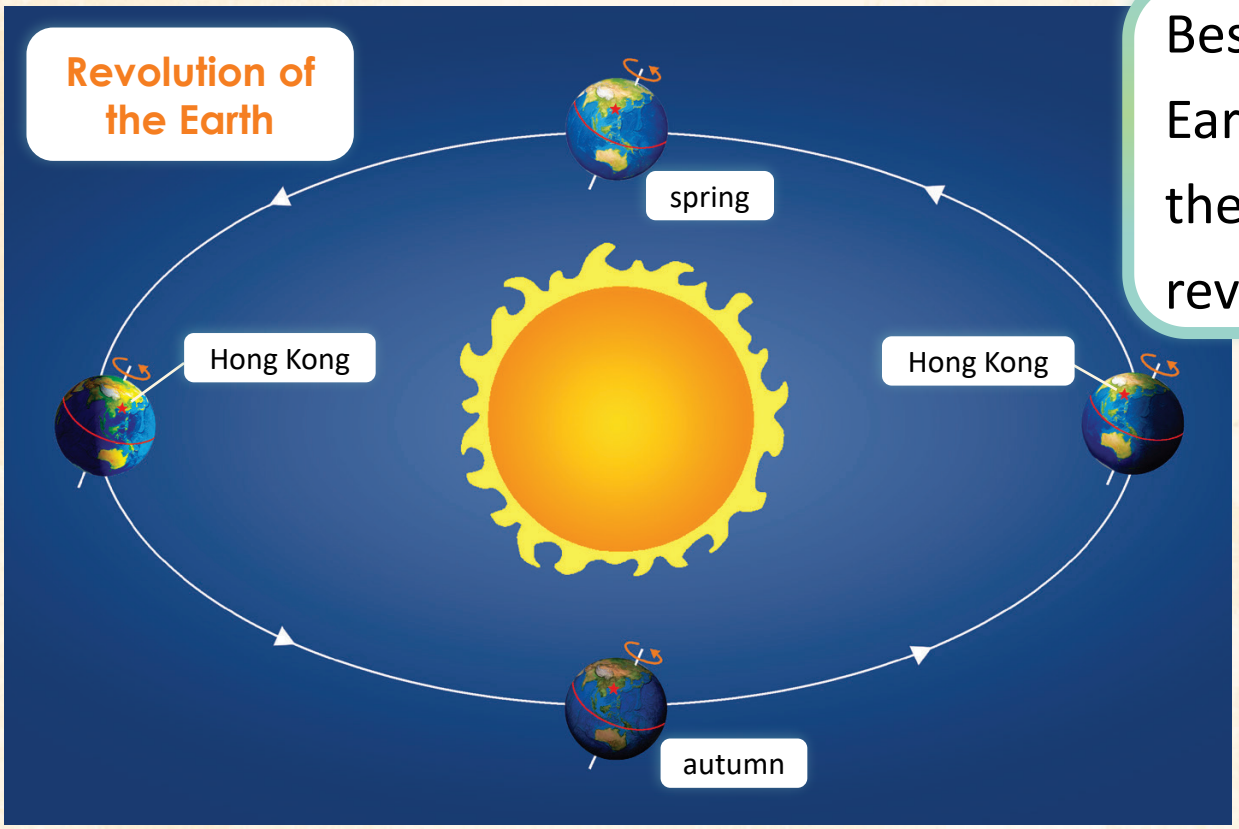


Quick Check

At different times of the day, shadows formed in the sunlight are the same / different in terms of length and direction.

5. A complete revolution takes a year, which is about 365 days.

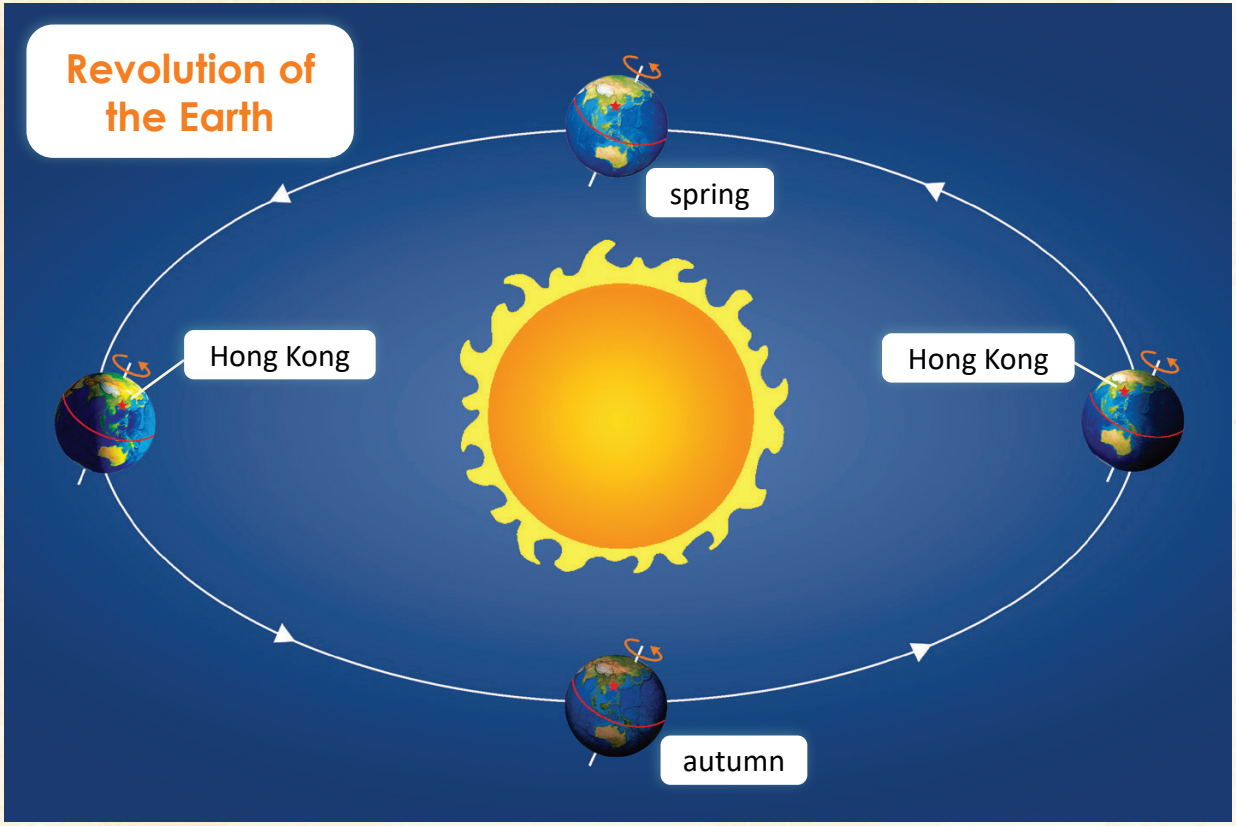
Look at the picture 'Revolution of the Earth' and write the correct answers on the _____.



Besides rotation, the Earth also moves around the Sun. This is called revolution.

5. A complete revolution takes a year, which is about 365 days.

Look at the picture ‘Revolution of the Earth’ and write the correct answers on the _____.



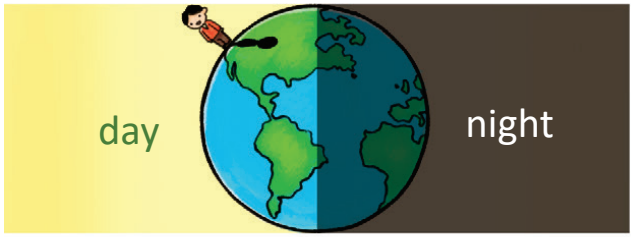
A complete revolution takes _a year_, which is about 365 days.

6. Since the Earth's axis is tilted, different places receive sunlight at different angles at different times of the year. This results in the changes of seasons.

Analyse the relationship between the angle of sunlight and the **changes of seasons** in the two hemispheres. Circle the correct answers.

Sunlight on the Earth / Item for comparison		
1. How the northern hemisphere is shone	The northern hemisphere receives slanted / overhead sunlight.	The northern hemisphere receives slanted / overhead sunlight.
2. Temperature	Temperature on the northern hemisphere is higher / lower ; temperature on the southern hemisphere is higher / lower	Temperature on the northern hemisphere is higher / lower ; temperature on the southern hemisphere is higher / lower
3. Season	The northern hemisphere: summer / winter ; the southern hemisphere: summer / winter	The northern hemisphere: summer / winter ; the southern hemisphere: summer / winter

I have learned



Rotation and revolution of the Earth

rotation → **Day and night**

revolution → **Changes of seasons**

